you may still view the data. The data is lost if one more disc dies, and until the failed disk(s) are repaired, NAS data recovery is not feasible.

The replacement of the incorrect disc by a user is another frequent error. Every time a disc is replaced, the whole NAS's data is rebuilt. The rebuild does not take place correctly if the wrong disc is replaced. In addition, the NAS RAID driver may become confused, which might result in more significant data loss.

## How to recover data from a RAID-based NAS?

Several factors could cause a RAID-based NAS to malfunction:

- Power outage or unexpected power spike
- improper shutdowns and overheating
- Attack by a virus or malware
- Error in setup or user action
- OS mistakes
- Faulty hardware controller
- Failure of the system, several discs, or both in rare instances physical injury

Without formatting the installed hard drives, a RAID-based NAS that is damaged or corrupt cannot be accessed or repaired. To get the damaged RAID-based NAS storage device working again, everything has to be reconfigured. This is a huge tragedy because the setup and any data on the RAID-based NAS storage were both lost.

In this section, we'll show you how to access your RAID-based NAS and retrieve data from it before formatting, resetting, or changing the settings of your RAID-based NAS.

### Steps to follow:

1. Turn off the RAID NAS, then remove all of its hard discs. Keep in mind how they are arranged in the rack.
2. Use USB or SATA to connect every NAS RAID drive to a Windows computer. Do not forget to connect all drives simultaneously.
3. Set up the system using Stellar Data Recovery Technician and start it up.
4. Decide whether to recover documents, emails, music, video, photographs, or other items and folders.
5. Select 'Next'. From the "Select Location" screen, select "RAID Recovery," and then click "Scan."
6. In the 'RAID Reconstruction' box, select the RAID 0, 5, or 6 tab, depending on the RAID NAS setup.
7. Using the left and right arrow buttons, move the RAID discs specified in the RAID list box to the "Move hard drives up/down for disc order" list box.
8. To arrange discs in the right sequence as they were in the RAID stack, use the up-down arrow buttons next to the list box.
9. Choose 'Don't know start sector of drives. If you are unsure about the disc order, display a list of possible start sectors. Then, choose one or more potential start sectors from the checklist box that is displayed, or, if none are listed there, use the Add Sector button to manually enter and add potential start sectors (a maximum of 32 are permitted).
10. From the drop-down menu, choose the "Stripe/Block size" and then click "Build RAID."
11. If the parameters supplied are accurate, the programme builds a likely RAID construction, and the produced RAID volumes are presented in the 'Select produced RAID' window. Activate the 'Show Volume List' button.
12. Select the volume from which data is to be retrieved in the 'Select Volume to Recover Data' boxes.
13. Select RAID volume and then click "Scan."
14. To start the 'Deep Scan' procedure if the necessary files are not detected in the scan results, click on the blue 'Click Here' link next to the word 'Deep Scan' at the bottom of the programme window.
15. All recovered folders and files are presented in the left 'Tree View' window. To discover, preview, and recover particular files or folders, use the search box in the top-right corner of the page. As an alternative, you may quickly explore and retrieve the necessary data by selecting the "File Type" option.
16. Select the files or folders you wish to recover by checking their respective boxes. Then select "Recover" from the menu. There is a pop-up.
17. To choose a specific place for data storing, click the 'Browse' option.
    *Note: Be careful not to choose the disc from the RAID NAS array that is currently attached to your PC. Choose a hard disc with suitable storage capacity, either an internal or external HDD.*
18. To save (recover) the chosen files and folders in the designated place, click the 'Start Saving' button.
19. Start your RAID NAS storage reconfiguration by formatting the discs when all data has been recovered. To use your NAS storage as before, copy the restored data to RAID NAS.
20. Whether your vital data is on your computer or a NAS RAID storage device, you should always make a backup of it. Backups speed up the recovery process and prevent you from losing crucial data and documents.

It's essential to establish a backup strategy that fits your needs, including both local and offsite backups. This way, you can quickly recover data in case of a disaster or hardware failure. Furthermore, keeping your NAS system secure by updating passwords, enabling firewalls, and implementing access controls is paramount to safeguarding your data from potential cyber threats.

Overall, by following the best practices and tips mentioned in this chapter, you can keep your NAS system running smoothly and securely, providing reliable and accessible storage for your data. Remember, regular maintenance is the key to keeping your NAS system running optimally and protecting your valuable data.